Mathematical Literature Research Database

Proofs, bounds, methods, and research connections

An Overleaf-ready typeset conversion of the supplied Excel workbook

Corpus	6 papers; 9 documented cross-paper connections; 6 research opportunities.
Source and scope	Scope: six supplied PDFs only. Mathematical expressions are stored as searchable LaTeX/plain-text notation; internal discrepancies and cross-model comparisons are explicitly flagged.
Reading convention	Each paper is divided into context, historical-versus-new bounds, upper-bound technique and proof, lower-bound technique and proof, and implications. Mathematical snapshots are typeset separately for readability.
Verification boundary	This is a faithful restructuring of the uploaded workbook, not an independent audit against the original papers. Statements marked Needs Verification in the workbook retain that qualification.

Paper directory

ID	Paper title	Year
P001	Quantum Algorithms for Learning and Testing Juntas	2007
P002	Testing Fourier Dimensionality and Sparsity	2011
P003	Quantum Learning of Concentrated Boolean Functions	2022
P004	Testing and Learning Quantum Juntas Nearly Optimally	2023
P005	A Brief Introduction to Quantum Query Complexity	2025
P006	Exponential Quantum Advantage in Testing Fourier Dimensionality	2026

I. Paper Index: Results and Proof Architecture

1 P001 — Quantum Algorithms for Learning and Testing Juntas

1. Research setting

Paper	Quantum Algorithms for Learning and Testing Juntas
Authors / year	Alp Atıcı; Rocco A. Servedio (2007)
Research problem	Test and PAC-learn Boolean k-juntas $f : \{-1,1\}^n \to \{-1,1\}$ under the uniform distribution using Fourier-sampling (FS) / uniform quantum-example access; compare with classical membership-query and quantum-membership-query models.

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	Testing: $O\big((k\log k)^2/\varepsilon\big)$ classical membership queries (Fischer et al.; paper Thm. III.1, p.5). Learning: $\widetilde{O}\big(n\cdot 2^{6k}\cdot\varepsilon^{-8}\big)$ uniform quantum examples via prior DNF learning (Thm. IV.2, p.11).	Testing: $O(k/\varepsilon)$ FS queries (Thm. III.3, p.6). Learning: $O((k\log k)/\varepsilon)$ FS queries plus $O\big(2^k\log(1/\varepsilon)\big)$ uniform classical examples (Thm. IV.5, pp.11–14).
Lower bound	Testing: $\Omega(k)$ classical membership queries for constant farness (Chockler–Gutfreund; Thm. III.2, p.5). Learning: $\Omega\big(2^k+\log n\big)$ classical membership queries (Lemma IV.1, p.10) and $\Omega\big(2^k\big)$ quantum membership queries at $\varepsilon = 1/10$ (Fact IV.3, p.11). These are different access models from FS.	Testing: $\Omega\Big(\sqrt{k}\Big)$ queries for FS-only testers (Thm. III.5, pp.7–10). Related learning restriction: with at most $2^k/101$ classical membership queries, $\Omega\big(2^k\big)$ quantum membership queries are needed (Obs. IV.8, p.14); this is not a matching FS lower bound.

Core bounds (typeset)	$$Q^{\mathrm{FS}}_{\mathrm{test}} = O(k/\varepsilon)$$ $$Q^{\mathrm{FS}}_{\mathrm{test}} = \Omega(\sqrt{k})$$ $$Q^{\mathrm{FS}}_{\mathrm{learn}} = O(k\log k/\varepsilon)$$ $$N_{\text{classical examples}} = O(2^k\log(1/\varepsilon))$$

3. Upper bound: method and proof

Method tags	Fourier sampling; support union; coupon collector; influence thresholding

Method	Each FS outcome S is drawn with probability $\lvert\widehat{f}(S)\rvert^2$. The tester accumulates variables in sampled supports; the learner first identifies variables with noticeable influence and then estimates the reduced truth table from uniform examples.
Core equations	$\Pr[S] = \lvert\widehat{f}(S)\rvert^2$ $T = 10(k+1)/\varepsilon$
Proof sketch	Testing: take $10(k+1)/\varepsilon$ FS samples and reject if their support union contains more than k variables. If f is ε -far from every k-junta, at least ε Fourier mass lies outside every current $\leq$ k-variable set, so the expected waiting time for each new variable is $\leq 1/\varepsilon$; Markov's inequality yields constant soundness. Learning adds influence filtering and a coupon-collector argument on relevant assignments.

4. Lower bound: method and proof

Method tags	Addressing function; hard distributions; birthday paradox; transcript indistinguishability
Method	The hard instances hide address variables so that fewer than order $\sqrt{k}$ samples are unlikely to reveal the collisions that distinguish the two distributions.
Core equations	$N = o(\sqrt{k}) \implies$ few observed collisions
Proof sketch	Apply Yao-style distributional reasoning to two addressing-function distributions. For $N = o(\sqrt{k})$, birthday-paradox bounds show that repeated addressee information is rare; conditioned transcripts therefore have small statistical distance, preventing bounded-error distinction.

5. Interpretation and traceability

Main improvement	Introduces a direct $O(k/\varepsilon)$ FS tester and a two-stage learner whose dimension dependence is near the information needed to expose relevant variables; leaves a $\sqrt{k}$-to-k testing gap.
Key insight	Distance from every k-junta forces Fourier weight outside every set of k variables. FS converts that spectral mass into observable support-growth events.
Research relevance	Direct foundation for quantum-example/FS junta testing and learning; supplies the explicit gap and the support-growth template used by later work.
Source references	Thm. III.1–III.5, pp.5–10; Lemma IV.1, Thm. IV.2, Fact IV.3, Thm. IV.5, Obs. IV.8, pp.10–14; conclusion p.14.

2 P002 — Testing Fourier Dimensionality and Sparsity

1. Research setting

Paper	Testing Fourier Dimensionality and Sparsity
Authors / year	Parikshit Gopalan; Ryan O'Donnell; Rocco A. Servedio; Amir Shpilka; Karl Wimmer (2011)
Research problem	Classically test whether a Boolean function $f : \mathbb{F}_2^n \to \{-1, 1\}$ has Fourier support contained in a k-dimensional subspace, and whether it has at most s nonzero Fourier coefficients.

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	For the exact same k-dimensional testing task, no earlier upper bound is isolated as the primary comparison in the supplied paper. Prior related algorithms concern degree, linearity, and sparse Fourier recovery.	k-dimensionality: $O(k \cdot 2^{2k}/\varepsilon)$ nonadaptive queries (informal Thm. 1, p.3; formal Thm. 10, p.13). s-sparsity: $O(s^6 \log s/\varepsilon^2 + s^14 \log s)$ nonadaptive queries (Thm. 8, pp.10–11). Induced-subclass implicit learning: $\text{poly}(2^k, 1/\varepsilon)$ queries (Thm. 18, p.25).
Lower bound	Fourier dimensionality: $\Omega(2^{k/2})$ adaptive queries was already known for constant-distance testing (Buhrman et al., quoted as Thm. 6); nonadaptive degree-k testing has $\Omega(2^k)$ lower bounds. Model/property distinctions matter.	k-dimensionality: $\Omega(2^{k/2})$ adaptive queries for (1/2- τ)-testing (Thm. 19, pp.26–27). Consequently, Fourier-sparsity testing has $\Omega(\sqrt{s})$ lower bounds for corresponding parameter choices.

Core bounds (typeset)	$Q_{\text{dimension}} = O(k2^{2k}/\varepsilon)$ $Q_{\text{sparsity}} = O(s^6 \log s/\varepsilon^2 + s^{14} \log s)$ $Q_{\text{dimension}} = \Omega(2^{k/2})$

3. Upper bound: method and proof

Method tags	Invariance subspace; random shifts; Fourier hashing; granular coefficients; implicit learning
Method	The algorithm detects the codimension of $\text{Inv}(f) = \{h : f(x+h) = f(x)\}$ using random shifts and sampled evaluations. Sparsity testing hashes Fourier coefficients into random-subspace buckets and estimates bucket weights.
Core equations	$\text{Inv}(f) = \{h : f(x+h) = f(x)\ \forall x\}$ $\dim \text{span}(\text{supp}\ \widehat{f}) = \text{codim}\ \text{Inv}(f)$

Proof sketch	For dimensionality, sample $\mathrm{O}(2^k)$ shifts h and test each candidate invariance relation on $\mathrm{O}(k \cdot 2^k/\varepsilon)$ random x values; enough violations certify that $\mathrm{Inv}(\mathrm{f})$ has excessive codimension. For sparsity, random subspaces isolate heavy coefficients, and a structural soundness theorem turns concentrated bucket mass into proximity to an s-sparse Boolean function.

4. Lower bound: method and proof

Method tags	Yao minimax; random coset structures; random Boolean functions; total-variation bounds
Method	Yes-instances are random functions constant on cosets of a random codimension-k subspace; no-instances behave like random Boolean functions conditioned to be far. Few adaptive queries reveal too little coset-collision information.
Core equations	$T = o(2^{k/2}) \implies$ few random-coset collisions
Proof sketch	Analyze deterministic decision trees under the two distributions. Until roughly 2^{k/2} queries, collisions among queried cosets remain unlikely, so answer transcripts have small total-variation distance. Yao's principle converts this to an adaptive randomized lower bound.

5. Interpretation and traceability

Main improvement	Provides the first broad property-testing framework in the paper for Fourier dimensionality and sparsity, including matching exponential-in-k lower-bound scale up to polynomial factors for dimensionality.
Key insight	Fourier dimension is the codimension of an invariance subspace, so spectral support geometry can be tested through time-domain shift invariance.
Research relevance	Baseline for Fourier-dimensionality testing and the main classical comparator for later quantum and improved classical algorithms.
Source references	Thm. 1 p.3; Thm. 6 (prior lower bound); Thm. 8 pp.10–11; Thm. 10 p.13; Thm. 18 p.25; Thm. 19 pp.26–27.

3 P003 — Quantum Learning of Concentrated Boolean Functions

1. Research setting

Paper	Quantum Learning of Concentrated Boolean Functions
Authors / year	Krishna Palem; Duc Hung Pham; M. V. Panduranga Rao (2022)
Research problem	Learn Boolean functions whose Fourier spectrum has M coefficients carrying at least 1- ε total squared mass. Study support recovery and PAC/exact learning with uniform quantum examples; use α for PAC error to distinguish it from spectral tail ε .

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	Classical comparisons include Kushilevitz–Mansour and sparse-FFT methods. Hassanieh et al. give $O(M \log(2^n/M) \cdot n/\varepsilon)$ time/query in the paper's Table 2 / Thm. 6 discussion. The paper's Thm. 5 statement and proof/table disagree on the precise KM-derived polynomial; exact transcription needs verification.	Support recovery: $O(M/\varepsilon)$ uniform quantum queries and output list size $O(M/\varepsilon)$ (Thm. 2, pp.12–13). PAC learning: $O(M/\varepsilon^2)$ uniform quantum queries, runtime $O(M^2/\varepsilon^3)$, and $\alpha = O(\varepsilon)$ (Thm. 3, p.14).
Lower bound	No directly matching prior quantum lower bound for this concentrated class is established in the supplied source. General classical/quantum learning bounds are discussed but use different access or distribution assumptions.	PAC learning under arbitrary distributions: $\Omega(M)$ quantum queries for $\alpha \leq 1/10$ (Thm. 9, p.18). Exact learning from uniform quantum examples: $\Omega(\varepsilon \cdot 2^n \cdot \log M/n)$ queries (Thm. 10, pp.18–21).

Core bounds (typeset)

$$Q_{\text{support}} = O(M/\varepsilon)$$

$$Q_{\text{PAC}} = O(M/\varepsilon^2)$$

$$Q_{\text{PAC}} = \Omega(M)$$

$$Q_{\text{exact}} = \Omega(\varepsilon 2^n \log M/n)$$

3. Upper bound: method and proof

Method tags	Quantum Fourier sampling; heavy-coefficient recovery; coefficient estimation
Method	Repeated QFS samples reveal coefficients with magnitude at least $\sqrt{\varepsilon/M}$. The recovered support is combined with sample-based coefficient/sign estimation to construct a low-error hypothesis.

Core equations	$$\sum_{S \in H} \lvert\widehat{f}(S)\rvert^2 \geq 1 - \varepsilon$$ $$\lvert\widehat{f}(S)\rvert \geq \sqrt{\varepsilon/M} \implies \Pr[S] \geq \varepsilon/M$$
Proof sketch	Any coefficient above threshold has sampling probability at least ε /M, so O(M/ε) samples find all such coefficients with constant success after amplification/log bookkeeping stated in the theorem. Truncating to the recovered heavy set controls the omitted Fourier mass; estimated coefficients then give an $\alpha = O(\varepsilon)$ PAC hypothesis.

4. Lower bound: method and proof

Method tags	VC dimension; information theory; mutual information; packing of perturbed subspace indicators
Method	The PAC lower bound follows from VC dimension at least M. The exact-learning lower bound builds a large family of concentrated functions whose uniform quantum-example states carry limited information per query.
Core equations	$$Q_{\text{PAC}} = \Omega(M)$$ $$Q_{\text{exact}} = \Omega(\varepsilon 2^n \log M/n)$$
Proof sketch	For PAC learning, embed an M-dimensional shattered family and invoke quantum sample-complexity lower bounds. For exact learning, encode many hypotheses by perturbing subspace-indicator constructions, bound per-copy mutual information, and use information-theoretic decoding requirements to obtain $\Omega(\varepsilon \cdot 2^n \cdot \log M/n)$.

5. Interpretation and traceability

Main improvement	Shows that QFS removes the ambient-dimension factor from dominant-support discovery and supplies both distribution-free PAC and exact uniform-example lower bounds.
Key insight	A concentrated spectrum is a probability distribution with few heavy atoms; quantum Fourier sampling accesses that distribution directly, while exact identification still requires resolving phase/sign information.
Research relevance	Directly relevant to quantum learning under Fourier concentration and to separating support identification from full function reconstruction.
Source references	Thm. 1 context; Thm. 2 pp.12–13; Thm. 3 p.14; Table 2 / Thm. 5–6 discussion; Thm. 7–10 pp.18–21. Needs Verification: exact KM bound due to an internal statement/table discrepancy.

4 P004 — Testing and Learning Quantum Juntas Nearly Optimally

1. Research setting

Paper	Testing and Learning Quantum Juntas Nearly Optimally
Authors / year	Thomas Chen; Shivam Nadimpalli; Henry Yuen (2023)
Research problem	Given query access to an n-qubit unitary U and $U^\dagger$, test whether U acts nontrivially on at most k qubits and learn an ε -close k-junta unitary under normalized Frobenius distance up to global phase.

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	Testing: Wang's quantum-junta tester uses $O(k)$ queries (originally $O(k \log k)$, improved by the Atıcı–Servedio analysis; p.4). No prior unitary-junta learning upper bound is isolated as matching the paper's model.	Testing: $O\left(\sqrt{k} \cdot \log^2 k/\varepsilon\right)$ calls as displayed in Thm. 20 (p.13), with polylog factors hidden in the abstract's $\widetilde{O}\left(\sqrt{k}\right)$. Learning: $O(k/\varepsilon + 4^k/\varepsilon^2)$ queries in Thm. 28 (pp.21–23); the proof text presents an $O(k \log k/\varepsilon^2)$ support-finding term before the dominant tomography term.
Lower bound	Testing inherits a $\sim\Omega\left(\sqrt{k}\right)$ Boolean-junta lower bound through reduction. For learning, the paper notes that a formal prior lower bound for this exact task was absent; generic process-tomography costs do not by themselves establish the claimed model-specific lower bound.	Testing: $\Omega\left(\sqrt{k}\right)$ queries (Thm. 24, pp.18–20). Learning: $\Omega(4^k \cdot \log(1/\varepsilon)/k)$ queries (Thm. 31, pp.25–26).

Core bounds (typeset)	$Q_{\text{test}} = O(\sqrt{k}\log^2 k/\varepsilon)$ $Q_{\text{test}} = \Omega(\sqrt{k})$ $Q_{\text{learn}} = O(k/\varepsilon + 4^k/\varepsilon^2)$ $Q_{\text{learn}} = \Omega(4^k \log(1/\varepsilon)/k)$

3. Upper bound: method and proof

Method tags	Qubit influence; Pauli sampling; influence estimation; quantum gapped group testing; Choi-state tomography
Method	The tester estimates group influences and applies quantum gapped group testing to locate evidence of more than k active qubits. The learner samples Pauli labels to identify influential qubits, then performs pure-state tomography on the corresponding Choi state.

Core equations	$Q_{\text{test}} = \widetilde{O}(\sqrt{k}/\varepsilon)$ $Q_{\text{tomography}} = O(4^k/\varepsilon^2)$
Proof sketch	Separate far-from-junta unitaries into cases captured by large outside influence or by many moderate influences. Influence-Estimator supplies a group-testing oracle, reducing support discovery to $\widetilde{\mathrm{O}}\left(\sqrt{k}\right)$ calls. For learning, restrict the Choi state to the discovered k-qubit subsystem and invoke pure-state tomography, producing the $4^k/\varepsilon^2$ dominant term.

4. Lower bound: method and proof

Method tags	Reduction from Boolean juntas; diagonal unitaries; communication complexity; metric packing; Input Guessing
Method	The testing reduction maps f to $U_f = \mathrm{diag}((-1)^{f(x)})$, transferring Boolean farness to unitary Frobenius distance. The learning lower bound packs many k-qubit unitaries and turns a learner into a low-communication decoder.
Core equations	$U_f = \mathrm{diag}((-1)^{f(x)})$ $\lvert\mathcal{N}_\varepsilon\rvert = \Omega\big((1/\varepsilon)^{4^k}\big)$
Proof sketch	Proposition 25 relates Boolean ε -farness to $\sqrt{\varepsilon/2}$-farness of diagonal unitaries, so the Boolean $\Omega\left(\sqrt{k}\right)$ bound applies. For learning, choose an ε -packing of size $\Omega\left((1/\varepsilon)^{4^k}\right)$; simulate each oracle query with $\mathrm{O}(k)$ communicated qubits and reduce to Input Guessing, forcing $\Omega\big(4^k \log(1/\varepsilon)/k\big)$ queries.

5. Interpretation and traceability

Main improvement	Obtains nearly matching k-dependence for testing and an essentially optimal 4^k-dimensional learning cost up to k and logarithmic factors.
Key insight	Pauli coefficients play the role of a Fourier spectrum for operators: their marginal mass defines qubit influence, while the Choi isomorphism converts unitary learning into state tomography.
Research relevance	Core quantum-operator analogue of Boolean junta testing/learning; connects Fourier/Pauli support discovery with tomography and communication lower bounds.
Source references	Thm. 20 p.13; Thm. 24 and Prop. 25 pp.18–20; Thm. 28 pp.21–23; Thm. 31 pp.25–26; future-work discussion p.5. Needs Verification: lower-order term/ ε presentation around Thm. 28.

5 P005 — A Brief Introduction to Quantum Query Complexity

1. Research setting

Paper	A Brief Introduction to Quantum Query Complexity
Authors / year	Yassine Hamoudi (2025)
Research problem	Tutorial survey of bounded-error quantum query complexity and its principal lower-bound methods, with examples and matching upper-bound constructions.

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	Not applicable as a single prior-upper comparison: this is a tutorial synthesizing existing algorithms and bounds rather than introducing one target problem.	No original new upper bound claimed. Representative surveyed results include $O(\sqrt{n})$ for OR through the adversary dual (Prop. 66, p.38), $\Theta(\sqrt{n})$ for AND–OR (Prop. 67), and $O(n^{3/2})$ for connectivity (Prop. 68).
Lower bound	Not applicable as a single prior-lower comparison: the document explains established lower-bound frameworks and examples.	No original new lower bound claimed. Representative derivations include $\Omega(\sqrt{n})$ for OR by the hybrid method and $Q_{\mu_{\text{unif}}}(\text{Search}) \geq \sqrt{m/15} - \sqrt{1/5}$ (Prop. 42, p.25).

Core bounds (typeset)	$Q(\text{OR}_n) = \Theta(\sqrt{n})$ $Q(\text{connectivity}) = O(n^{3/2})$

3. Upper bound: method and proof

Method tags	Adversary SDP dual; span-program-style constructions; surveyed quantum algorithms
Method	The upper-bound side uses feasible solutions to the dual adversary semidefinite program to certify query algorithms and compose solutions.
Core equations	$\text{Adv}^{\pm}(f)$ controls bounded-error quantum queries
Proof sketch	Translate a function into an adversary-dual feasibility problem, build vectors satisfying the pairwise constraints, and read query complexity from the objective. Worked examples recover standard square-root speedups and composition behavior.

4. Lower bound: method and proof

Method tags	Hybrid argument; polynomial method; recording/compressed oracle; adversary method
Method	Each method limits how fast queries can separate states, raise approximation degree, accumulate database information, or satisfy adversary progress constraints.
Core equations	$Q_{\mu_{\text{unif}}}(\text{Search}) \geq \sqrt{m/15} - \sqrt{1/5}$
Proof sketch	Choose a hard input distribution or progress measure, bound the change caused by one query, and telescope over T queries. Alternative sections encode acceptance probabilities as low-degree polynomials, record queried information coherently, or optimize an adversary matrix/SDP.

5. Interpretation and traceability

Main improvement	Pedagogical synthesis rather than a new asymptotic result; useful as a method-selection reference and source of reusable proof templates.
Key insight	Quantum query lower bounds can be viewed through several equivalent or complementary bottlenecks: state distinguishability, polynomial approximation, recorded information, and adversary progress.
Research relevance	Methodological reference for designing or auditing lower bounds in junta and Fourier-structure problems; not direct evidence for a new task-specific bound.
Source references	Secs. 2–6; Prop. 42 p.25; Prop. 51 p.32; Prop. 66–68 p.38 and following pages.

6 P006 — Exponential Quantum Advantage in Testing Fourier Dimensionality

1. Research setting

Paper	Exponential Quantum Advantage in Testing Fourier Dimensionality
Authors / year	Kenny Chen (2026)
Research problem	Test whether a Boolean function $f : \mathbb{F}_2^n \to \{\pm 1\}$ has Fourier dimension at most k or is ε -far, under classical and quantum query access.

2. Historical comparison: previous versus current results

Type	Earlier literature	This paper
Upper bound	Classical: Gopalan et al. give $\mathrm{O}\big(k \cdot 2^{2k}/\varepsilon\big)$; Alekseychuk–Konyushok improve this to $\mathrm{O}\big(k^2 \cdot 2^k/\varepsilon\big)$ (paper introduction, p.3). No prior quantum upper with the paper's exponential improvement is identified.	Quantum: one-sided $\mathrm{O}(k/\sqrt{\varepsilon})$ queries (Thm. 3.1, pp.9–10). Classical: $\mathrm{O}\big(k \cdot 2^{k/2}/\varepsilon\big)$=$\widetilde{\mathrm{O}}\big(2^{k/2}/\varepsilon\big)$ queries (Thm. 4.1, pp.14–18).
Lower bound	Classical: $\Omega\big(2^{k/2}\big)$ queries from Gopalan et al. For quantum testing, no matching prior lower is identified in the supplied source.	Quantum: $\Omega(k)$ queries for fixed $\varepsilon < 1/4$ (Thm. 3.2, pp.12–14), matching the upper bound in k for constant ε . No new classical lower bound beyond the cited $\Omega\big(2^{k/2}\big)$.

Core bounds (typeset)	$Q^{\text{old}}_{\text{classical}} = O(k2^{2k}/\varepsilon)$ $Q^{\text{new}}_{\text{classical}} = O(k2^{k/2}/\varepsilon)$ $Q_{\text{quantum}} = O(k/\sqrt{\varepsilon})$ $Q_{\text{quantum}} = \Omega(k) \quad (\varepsilon < 1/4)$

3. Upper bound: method and proof

Method tags	Quantum Fourier sampling; amplitude amplification; iterative span growth; collision fingerprints
Method	The quantum tester maintains the span W of observed Fourier samples and amplitude-amplifies mass outside W until it either finds $k+1$ independent vectors or certifies low dimension. The classical tester compares collision rates of evaluation fingerprints under random shifts.
Core equations	$2\delta_W(f) \le \mathrm{wt}_{\overline{W}}(f)$ $\mathrm{wt}_{\overline{W}}(f) \ge 2\varepsilon$ $Q_{\text{new vector}} = O(1/\sqrt{\varepsilon})$

Proof sketch	Lemma 3.2 shows $2\delta_U(f) \leq \mathrm{wt}_{\overline{U}}(f)$. If f is ε -far from every k-dimensional function, every span W of dimension $\leq$ k leaves at least 2 ε Fourier mass outside. A coherent Fourier-sampling procedure plus amplitude amplification finds a new independent vector in $O(1/\sqrt{\varepsilon})$ queries; $k+1$ rounds reject, while true k-dimensional functions never reject.

4. Lower bound: method and proof

Method tags	Reduction from Simon's problem; random masks; periodic-versus-injective promise
Method	Encode a Simon instance g through a random mask M$\circ$g. Periodic instances yield Fourier dimension at most k; injective instances induce a random Boolean function that is far from possessing any nonzero period with high probability.
Core equations	Simon quantum query lower bound $= \Omega(k)$
Proof sketch	Map the Simon promise to Fourier-dimensionality testing. A hidden XOR period constrains the Fourier support to a codimension-one subspace, whereas a random masked injective function is far from all such periodic functions. Simon's $\Omega(k)$ quantum lower bound then transfers.

5. Interpretation and traceability

Main improvement	Reduces the quantum complexity from exponential to linear in k for constant ε and improves the classical upper bound to the known exponential lower-bound scale up to polynomial factors.
Key insight	Distance from a Fourier subspace lower-bounds spectral mass outside it; coherent access can quadratically accelerate discovery of that outside mass.
Research relevance	Most direct current paper for quantum Fourier-dimensionality testing; provides the amplitude-amplified span-growth paradigm and a tight-in-k quantum lower bound.
Source references	Thm. 3.1 and Lemma 3.2 pp.9–10; Thm. 3.2 pp.12–14; Thm. 4.1 pp.14–18. Needs Verification: Thm. 3.2 appears to say "Fourier degree" once although context and proof concern Fourier dimension.

II. Connections and Research Opportunities

Evidence boundary	Evidence is limited to the six supplied PDFs. 'Inferred' and 'Speculative' entries are hypotheses for investigation, not established novelty claims.
Reading convention	Connections are represented individually to keep shared foundations, model differences, techniques and source references readable. Research opportunities preserve the original status and verification cautions.

7 Cross-paper connections

7.1 C001 — P001 / P003: Shared framework

Papers	P001 ⟷ P003
Shared ground	Both use uniform quantum examples / Fourier sampling, where outcome S has probability $\lvert\widehat{f}(S)\rvert^2$.
Technical caveat	P001 targets k-juntas; P003 targets spectra concentrated on M coefficients. P003 generalizes the sampling workflow but not the property.
Proof / method link	FS/QFS distribution; heavy support discovery
Evidence	P001 Thm. IV.5; P003 Thm. 2–3
Research use	Directly compare support discovery cost and the extra work needed to recover coefficient values/signs.
Priority	High

7.2 C002 — P001 / P004: Analogy / lift

Papers	P001 ⟷ P004
Shared ground	Both locate influential coordinates before learning a reduced object.
Technical caveat	Boolean variables/Fourier coefficients versus qubits/Pauli coefficients; P004 also requires unitary tomography.
Proof / method link	Influence; support discovery; reduced reconstruction
Evidence	P001 Thm. III.3/IV.5; P004 Thm. 20/28
Research use	Use the Boolean-to-operator analogy to separate generic support discovery from model-specific reconstruction.
Priority	High

7.3 C003 — P002 / P006: Direct improvement

Papers	P002 ⟷ P006
Shared ground	Same Fourier-dimensionality property over $\mathbb{F}_2^n$.
Technical caveat	P006 adds coherent quantum access and also improves the classical algorithm; ε and one-sidedness must be tracked.
Proof / method link	Invariance/subspace geometry; collision versus amplitude amplification
Evidence	P002 Thm. 10/19; P006 Thm. 3.1/3.2/4.1
Research use	Primary before/after line: $O\left(k2^{2k}/\varepsilon\right)$ to $O\left(k2^{k/2}/\varepsilon\right)$ classically and $O(k/\sqrt{\varepsilon})$ quantumly.
Priority	High

7.4 C004 — P001 / P002: Structural comparison

Papers	P001 ⟷ P002
Shared ground	Both test low-dimensional Fourier structure using queries.
Technical caveat	A k-junta restricts supports to subsets of k coordinates; Fourier dimension allows any k-dimensional linear subspace. Neither property is identical to the other.
Proof / method link	Coordinate support versus linear-span support
Evidence	P001 Sec. III; P002 definitions and Thm. 10
Research use	Prevents transferring bounds without checking whether the subspace is coordinate-aligned.
Priority	Medium

7.5 C005 — P004 / P005: Methodological support

Papers	P004 ⟷ P005
Shared ground	P004's lower bounds rely on distinguishability/reduction ideas covered by the query-complexity toolkit.
Technical caveat	P005 is a tutorial and does not prove P004's task-specific bounds; P004 uses communication packing beyond the tutorial examples.
Proof / method link	Reduction; adversary/hybrid viewpoint; information limits
Evidence	P004 Thm. 24/31; P005 Secs. 2–6

Research use	Use P005 to audit proof architecture and select candidate methods for stronger lower bounds.
Priority	Medium

7.6 C006 — P001 / P003: Class inclusion / extension

Papers	P001 $\longleftrightarrow$ P003
Shared ground	A Boolean k-junta has Fourier support on at most 2^k subsets, hence is a special structured concentrated/sparse case.
Technical caveat	P003's parameter M ignores which variables generate the support; translating $M = 2^k$ can lose junta structure.
Proof / method link	Sparsity/concentration; QFS support recovery
Evidence	P001 Fourier preliminaries; P003 definition and Thm. 2
Research use	Compare whether exploiting coordinate structure beats generic M-heavy-spectrum learning.
Priority	High

7.7 C007 — P003 / P006: Incomparable spectral measures

Papers	P003 $\longleftrightarrow$ P006
Shared ground	Both infer structure from the distribution $\lvert\widehat{f}(S)\rvert^2$.
Technical caveat	Top-M concentration counts heavy atoms; Fourier dimension measures the span of all nonzero support. A spectrum may be concentrated yet high-dimensional, or low-dimensional yet dense.
Proof / method link	Heavy atoms versus span dimension
Evidence	P003 definitions; P006 problem definition/Lemma 3.2
Research use	Suggests a two-parameter landscape indexed by tail mass and span dimension.
Priority	High

7.8 C008 — P001 / P006: Algorithmic refinement

Papers	P001 ⟷ P006
Shared ground	Both grow a learned support/span from Fourier samples until a dimension threshold is crossed.
Technical caveat	P001's FS oracle returns classical samples; P006's amplitude amplification needs coherent phase-query access and reflections. The quadratic ε improvement does not automatically transfer to FS-only access.
Proof / method link	Support growth; amplitude amplification; oracle separation
Evidence	P001 Thm. III.3/III.5; P006 Thm. 3.1
Research use	Isolate coherent access as the resource responsible for accelerating rare outside-mass discovery.
Priority	High

7.9 C009 — P003 / P004: Workflow analogy

Papers	P003 ⟷ P004
Shared ground	Both first identify a small relevant support and then reconstruct an object on it.
Technical caveat	P003 estimates scalar Fourier coefficients for a Boolean hypothesis; P004 performs pure-state tomography of a Choi state, causing 4^k dependence.
Proof / method link	Support recovery followed by reconstruction
Evidence	P003 Thm. 2–3; P004 Thm. 28
Research use	Clarifies which complexity comes from finding support and which comes from describing the reduced object.
Priority	Medium

8 Research opportunities

Opportunity overview

ID	Research direction	Evidence class
O001	Close the FS-only Boolean junta-testing gap	Explicit open problem
O002	Tolerant testing of dominant Fourier mass	Inferred research gap
O003	Amplitude-amplified support discovery beyond span testing	Speculative direction
O004	Sharper QEX lower bounds for concentrated-function learning	Inferred research gap
O005	Separate support recovery from exact sparse Boolean learning	Speculative direction
O006	Tolerant testing for quantum juntas and extension to channels	Explicit future direction

8.1 O001 — Close the FS-only Boolean junta-testing gap

Evidence class	Explicit open problem
Source papers	P001
Evidence / motivation	P001 leaves $O(k/\varepsilon)$ upper versus $\Omega\left(\sqrt{k}\right)$ lower for Fourier-sampling testers.
Research question	Can an FS-only tester use $o(k/\varepsilon)$ samples, or can the lower bound be strengthened toward k for constant ε ?
Direction	Information-theoretic
Candidate approach	Construct hard spectral distributions with matched low-order collision statistics; alternatively seek a support-size estimator that avoids discovering $k+1$ distinct variables.
Success criterion	An intermediate $\Omega\left(k^{1/2+\delta}\right)$ lower bound or $o(k)$ constant- ε tester would be meaningful.
Main obstacle	FS produces classical samples, so coherent amplitude amplification from P006 is unavailable.
Priority	High
Source references	AS07 conclusion p.14; Thm. III.3/III.5
Novelty / verification	Open in supplied corpus; present-day status requires external literature verification.

8.2 O002 — Tolerant testing of dominant Fourier mass

Evidence class	Inferred research gap
Source papers	P003; P006
Evidence / motivation	The corpus separately treats top-M concentration and low-dimensional span, but does not give a tolerant tester for quantitative top-k spectral mass.
Research question	Estimate or distinguish thresholds for M_k(f)=sum of the k largest $\lvert\widehat{f}(S)\rvert^2$ using quantum-example or coherent phase-query access.
Direction	Algorithmic + lower bound
Candidate approach	Combine heavy-hitter discovery with tolerant distribution testing; compare sample-only QFS to coherent amplitude-estimation access.
Success criterion	Upper/lower bounds parameterized by k, gap, and tail mass; explicit oracle-model separation.
Main obstacle	Coefficient signs are irrelevant for mass estimation but Boolean realizability and unknown support complicate lower bounds.
Priority	Medium
Source references	P003 Thm. 2–3; P006 Lemma 3.2/Thm. 3.1
Novelty / verification	Inferred from supplied papers; novelty/status not externally verified.

8.3 O003 — Amplitude-amplified support discovery beyond span testing

Evidence class	Speculative direction
Source papers	P001; P003; P006
Evidence / motivation	P006 quadratically accelerates discovery of Fourier mass outside a known subspace; P001/P003 rely on repeated sample-only observations.
Research question	Under coherent access, can outside-mass amplification improve junta or concentrated-spectrum support discovery in ε without losing correctness under approximate structure?
Direction	Algorithmic
Candidate approach	Build a reflection about the currently learned coordinate/subspace support, amplify its complement, and analyze errors as the support estimate evolves.
Success criterion	A rigorously specified access model and an ε -dependence improvement over repeated QFS for a nontrivial task.

Main obstacle	The reflection may require knowledge unavailable from a sample oracle; false support decisions can accumulate.
Priority	Medium
Source references	P001 Thm. III.3/IV.5; P003 Thm. 2; P006 Thm. 3.1
Novelty / verification	Speculative synthesis, not a claim of novelty.

8.4 O004 — Sharper QEX lower bounds for concentrated-function learning

Evidence class	Inferred research gap
Source papers	P003; P005
Evidence / motivation	P003 gives $\Omega(M)$ PAC and an information-theoretic exact-learning bound, leaving room to characterize dependence on ε , distribution, and the distinction between support and phase recovery.
Research question	What is the tight uniform-QEX complexity for PAC learning M-concentrated Boolean functions at error α and tail ε ?
Direction	Lower bound
Candidate approach	Use state discrimination, adversary matrices, or packing families that share nearly identical heavy supports but differ in coefficient signs.
Success criterion	A lower bound matching $\mathrm{O}\big(M/\varepsilon^2\big)$ in a clearly aligned model, or an improved algorithm showing the upper bound is loose.
Main obstacle	Boolean Fourier coefficients obey realizability constraints, limiting arbitrary state packings.
Priority	Medium
Source references	P003 Thm. 3/9/10; P005 lower-bound methods
Novelty / verification	Inferred from local corpus; current literature must be checked before claiming openness.

8.5 O005 — Separate support recovery from exact sparse Boolean learning

Evidence class	Speculative direction
Source papers	P003
Evidence / motivation	The corpus gives efficient heavy-support recovery but exact learning can remain exponentially hard.
Research question	For exactly M-sparse Boolean functions, which additional promises make signs/values recoverable with $\mathrm{poly}(M, n)$ quantum queries?

Direction	Structural + algorithmic
Candidate approach	Classify realizable sparse spectra, exploit convolution/idempotence constraints of Boolean functions, and test candidate supports before coefficient reconstruction.
Success criterion	A promise class with polynomial exact-learning complexity or a structural obstruction proving exponential cost.
Main obstacle	The supplied papers do not establish the full current state of exact sparse Boolean learning.
Priority	Low
Source references	P003 Thm. 2 and Thm. 10
Novelty / verification	Speculative; requires broader literature review.

8.6 O006 — Tolerant testing for quantum juntas and extension to channels

Evidence class	Explicit future direction
Source papers	P004
Evidence / motivation	P004 explicitly identifies tolerant property testing and quantum-channel extensions as future directions.
Research question	Can one distinguish distance $\leq \varepsilon_1$ from $\geq \varepsilon_2$ to k-junta unitaries, and can analogous support notions be tested for general quantum channels?
Direction	Algorithmic + lower bound
Candidate approach	Develop robust influence estimators under a gap promise; replace pure Choi states by mixed Choi operators and quantify how channel noise affects group testing.
Success criterion	Query bounds with explicit dependence on k, ε_2-ε_1, and channel rank/noise; matching reductions where possible.
Main obstacle	Mixed-state tomography and completely positive constraints may dominate support testing.
Priority	High
Source references	P004 future-work discussion p.5; Thm. 20/28
Novelty / verification	Explicit in supplied 2023 paper; status as of 2026 not externally verified.